Mohamed Ayman Mohamed

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Summary

AI research scientist and software engineer working on continual reinforcement learning, with experience spanning research (University of Alberta, KAUST, Microsoft) and production systems (AWS). Published at RLC, ICLR, and TPAMI; MSc with a 4.0 GPA. Built a continual RL environment and benchmark now used to study how agents cope with gradual and abrupt distribution shift. Interests span continual learning, reinforcement learning, representation learning, and distributed systems.

Research Experience

Graduate Research Assistant — Reinforcement Learning | *Full-time*

University of Alberta — Machado Lab Apr 2024 – Aug 2025

- Built a C++/Python continual reinforcement learning environment modelling **gradual** and **abrupt** distribution shifts in episodic and non-episodic settings, across **15** simulation scenarios.
- Benchmarked **DQN, PPO**, and **SAC** on policy stability, convergence speed, and sample efficiency, exposing a **90%** failure rate under continual learning conditions.
- Roughly doubled throughput — grid observations from **1,200** to **2,200** steps/second and screen observations from **800** to **1,600** — via CUDA rendering and asynchronous inference.
- Added quad-tree and precision-approximation structures to cut computational overhead, and a CNN feature-extraction pipeline learning compact state representations from high-dimensional inputs (**30%** faster convergence, **25%** better sample efficiency).
- Containerised with Docker behind a CI/CD pipeline of **25+** automated tests; resolved memory leaks with Valgrind and tracked regressions with Google Benchmark.

Machine Learning Research Intern | *Remote*

KAUST — King Abdullah University of Science and Technology Feb 2023 – Oct 2023

- Implemented a chain-of-thought ground-truth generation tool, improving 3D object detection accuracy by **8%** on complex visual grounding tasks.
- Built a rule-based anchors parser for automated object localisation — published at **ICLR 2024**.
- Engineered an NLP data pipeline using tokenisation, TF-IDF embeddings, and semantic filtering to parse and sort anchor descriptions, improving anchor–target matching precision by **15%**.
- Optimised **3DCoMPaT++** segmentation with **PointNet++** and diffusion-based refinement, deployed a full-server solution, and improved IoU by **10%**.

Undergraduate Research Assistant — Machine Learning | *Part-time*

American University in Cairo Sep 2022 – Apr 2023

- Optimised disk-based approximate nearest-neighbour (ANN) search by designing an inverted-index query execution engine, achieving a **6**× speedup over Microsoft's SPANN while maintaining recall and graph quality.
- Improved retrieval accuracy by **15%** using embeddings from autoencoders and transformer models; extracted **100k+** meaningful queries via autoencoder feature learning.
- Established automated pipelines for embedding preprocessing and ANN index construction, reducing end-to-end latency by **30%** on billion-scale datasets.
- Created a COVID-19 policy optimisation framework using LSTM forecasting, deep Q-networks, and graph neural networks for regional modelling.

Work Experience

Software Engineer | *Full-time*

Amazon Web Services Sep 2025 – Present

- Reduced AWS region build-model creation from **days to five minutes** by designing an AI-powered code generation system on Amazon Bedrock.
- Initiated an AI automation program for EKS operational workflows, targeting the highest-impact repetitive tasks: build-model authoring, pipeline failure triage, ticket resolution, and dashboard analysis.
- Architected cross-partition observability infrastructure and automated data pipelines that adapt to regional constraints, reducing operational overhead by **40%** and enabling zero-touch deployments.
- Owned end-to-end design and delivery of automated testing frameworks, regulatory compliance workflows, and a resilient resource cleanup API, reducing customer escalations by **25%**.

Software Engineer | *Contract*
University of Alberta Oct 2024 – Apr 2025

- Developed a distributed simulation module in C# and C++ adhering to the High-Level Architecture (HLA) standard, reducing communication latency by **20%**.
- Designed a scalable PostgreSQL database managing **500k+** records across **6** federates, enabling centralised performance analysis and real-time monitoring.
- Deployed an N-tier architecture in ASP.NET with PostgreSQL supporting **500+** concurrent user requests at sub-second response times.
- Embedded the CPython interpreter in a C# application to run Python analysis and machine learning workflows inside the .NET environment.

Applied Scientist Intern | *Full-time*
Microsoft — Zcode Team Jul 2022 – Sep 2022

- Developed a meta-learning data distillation algorithm reducing dataset size by **99.9%** while maintaining **95%** of model performance across transformer architectures.
- Benchmarked across multiple NLP datasets, achieving a **10%** improvement over the state of the art in machine translation, classification, and named-entity recognition.
- Automated **20** Python workflows for hyperparameter tuning and documented **5** optimisation procedures for the internal knowledge base.
- Demonstrated empirically across **10** controlled experiments that model non-linearity significantly affects performance on distilled data.

Software Engineer | *Full-time*
Suitera L.L.C. Sep 2021 – Apr 2022

- Identified bottlenecks in slab data processing for complex circuit simulations and multi-die package assemblies on Suitera's SuiEM platform.
- Designed C++ approximation algorithms for precise slab representation, using heuristic and iterative methods to simplify complex geometries while preserving essential structural features.
- Built aggregation techniques consolidating similar slab segments, significantly reducing computational overhead and simulation runtime.
- Validated with custom GoogleTest fixtures, profiled with gdb and Valgrind, and developed OpenGL 3D visualisations of circuit structure and current flow.

Full-Stack Software Engineering Intern | *Remote*
Envelope.network Inc. Jun 2021 – Aug 2021

- Architected a cross-browser extension increasing user engagement by **15%** through behavioural analysis of **1000+** browsing data points.
- Built a **5**-stage recommendation pipeline ranking content for **500+** users, reaching **0.76** precision on textual data.
- Built and maintained **10+** RESTful endpoints in Scala with Finatra, and integrated PostgreSQL via the Slick functional relational mapper.

Publications

The Cell Must Go On: Agar.io for Continual Reinforcement Learning
Mohamed A. Mohamed, Kateryna Nekhomiazh, Marcos M. José, Andrew Patterson, Marlos C. Machado
Reinforcement Learning Conference (RLC) 2026

Self-Referenced Social Preferences: Cooperation without Observing Others' Rewards
Mohamed Ayman Mohamed, Harshil Kotamreddy, Marcos Menon José
Preprint, arXiv:2610.07881 2026

CoT3DRef: Chain-of-Thoughts Data-Efficient 3D Visual Grounding
Eslam Mohamed Bakr, **Mohamed Ayman**, Mahmoud Ahmed, Habib Slim, Mohamed Elhoseiny
International Conference on Learning Representations (ICLR) 2024

3DCoMPaT++: An Improved Large-Scale 3D Vision Dataset for Compositional Recognition
Habib Slim, Xiang Li, Yuchen Li, Mahmoud Ahmed, **Mohamed Ayman**, et al.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2025

Education

MSc in Computer Science — Artificial Intelligence
University of Alberta Sep 2023 – Aug 2025
GPA: 4.0/4.0 | Supervisor: Prof. Marlos C. Machado | Awards: Travel Award 2024/25, Graduate Research Assistantship

BSc in Computer Engineering, Minor in Mathematics
American University in Cairo Sep 2018 – May 2023
GPA: 3.97/4.0 | Dean's List (4 consecutive years) | Al-Ghurair Scholarship ('18–'23) | Bronze Medal, ECPC '23; ACPC Finalist '23

Teaching

Graduate Teaching Assistant, University of Alberta: Fundamentals of Reinforcement Learning, Introduction to Machine Learning, Fundamentals of Databases, Introduction to Networks.

Undergraduate Teaching Assistant, American University in Cairo: Introduction to Machine Learning, Deep Learning in Practice, Analysis & Design of Algorithms, Fundamentals of Computing II.

Technical Skills

Programming Languages: Python, C/C++, C#, Java, Scala, JavaScript, Rust, SQL

ML/AI Frameworks: PyTorch, TensorFlow, PyTorch Lightning, JAX, Scikit-Learn, Hugging Face, XGBoost

RL Libraries: CleanRL, PFRL, Stable-Baselines3, RLlib, Gymnasium

NLP & CV: spaCy, NLTK, Transformers, OpenCV, PointNet++

Cloud & DevOps: AWS (CDK, Lambda, S3, EC2), Azure, Docker, Kubernetes, CI/CD

Low-level & Tools: CUDA, OpenMP, OpenGL, x86-64 Assembly, RISC-V, Verilog, CMake, Valgrind, Git